



EAST TEXAS A&M
— UNIVERSITY —

CSCI 352.61E

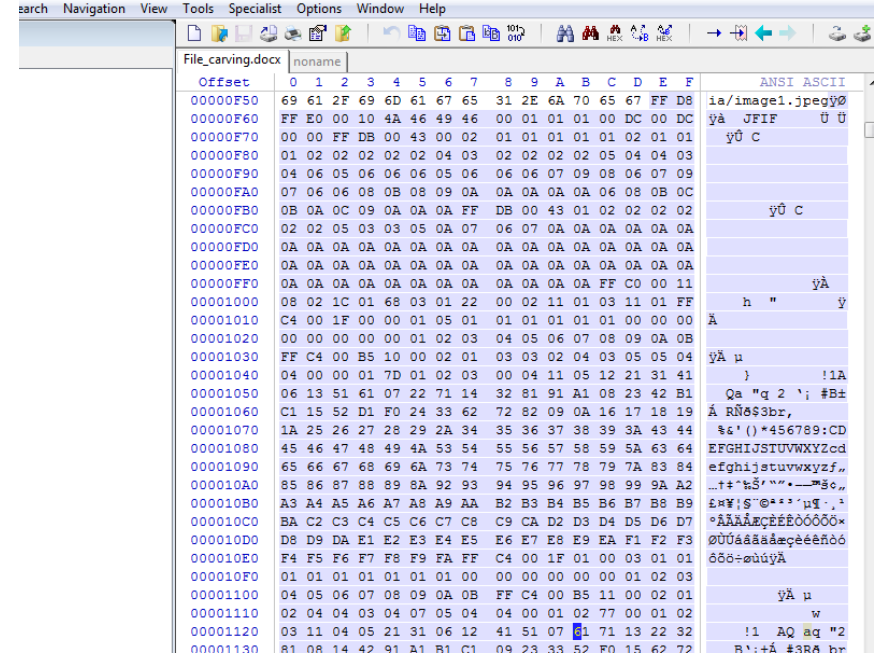
DIGITAL FORENSICS

FILE CARVING

- A well known computer forensics term
- To identify and extract (carve) the file
 - from unallocated clusters
 - without the assistance of the file system that originality created the file.
- How to extract files?
 - based on signature information alone.

AN EXAMPLE SCENARIO

- A suspect deleted a file
- The files contains an important image
- Image occupies a few clusters
- Unfortunately, some clusters are reused (Overwritten) by new files
- A forensic expert really wants to recover the image, even a partial image



FILE SIGNATURE

- x Referred to as a magic number,
- x Used to identify a file format
 - x a constant numerical or
 - x text value used to.

Digital Forensics Magic Numbers x +

← → ↻ 🏠 <https://asecuritysite.com/forensics/magic>

Digital Investigation

[Back] These pages outlines some digital forensics examples showing the magic number which identifies the file:

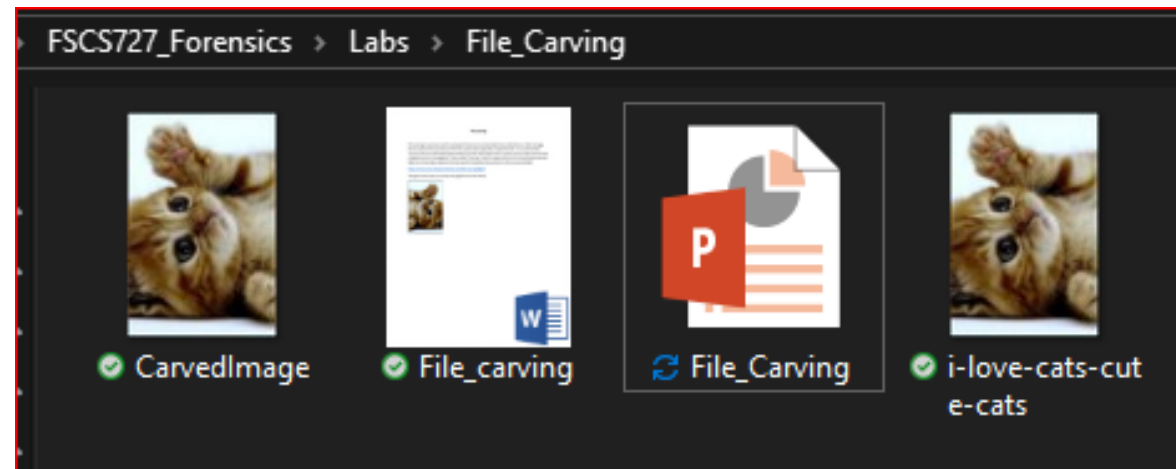
Description	Extension	Magic Number
Adobe Illustrator	.ai	25 50 44 46 [%PDF]
Bitmap graphic	.bmp	42 4D [BM]
Class File	.class	CA FE BA BE
JPEG graphic file	.jpg	FFD8
JPEG 2000 graphic file	.jp2	0000000C6A5020200D0A [....JP..]
GIF graphic file	.gif	47 49 46 38 [GIF89]
TIF graphic file	.tif	49 49 [II]
PNG graphic file	.png	89 50 4E 47 .PNG
Photoshop Graphics	.psd	38 42 50 53 [8BPS]
Windows Meta File	.wmf	D7 CD C6 9A
MIDI file	.mid	4D 54 68 64 [MThd]
Icon file	.ico	00 00 01 00
MP3 file with ID3 identity tag	.mp3	49 44 33 [ID3]
AVI video file	.avi	52 49 46 46 [RIFF]
Flash Shockwave	.swf	46 57 53 [FWS]
Flash Video	.flv	46 4C 56 [FLV]
Mpeg 4 video file	.mp4	00 00 00 18 66 74 79 70 6D 70 34 32 [....ftypmp42]
MOV video file	.mov	6D 6F 6F 76 [....moov]
Windows Video file	.wmv	30 26 B2 75 8E 66 CF
Windows Audio file	.wma	30 26 B2 75 8E 66 CF

VMWare Disk file	.vmdk	4B 44 4D 56 [KDMV]
Outlook Post Office file	.pst	21 42 44 4E 42 [!BDN]
PDF Document	.pdf	25 50 44 46 [%PDF]
Word Document	.doc	D0 CF 11 E0 A1 B1 1A
RTF Document	.rtf	7B 5C 72 74 66 31 [{
Excel Document	.xls	D0 CF 11 E0 A1 B1 1A
PowerPoint Document	.ppt	D0 CF 11 E0 A1 B1 1A
Visio Document	.vsd	D0 CF 11 E0 A1 B1 1A
DOCX (Office 2010)	.docx	50 4B 03 04 [PK]
XLSX (Office 2010)	.xlsx	50 4B 03 04 [PK]
PPTX (Office 2010)	.pptx	50 4B 03 04 [PK]
Microsoft Database	.mdb	53 74 61 6E 64 61 72
Postscript File	.ps	25 21 [%!]
Outlook Message File	.msg	D0 CF 11 E0 A1 B1 1A

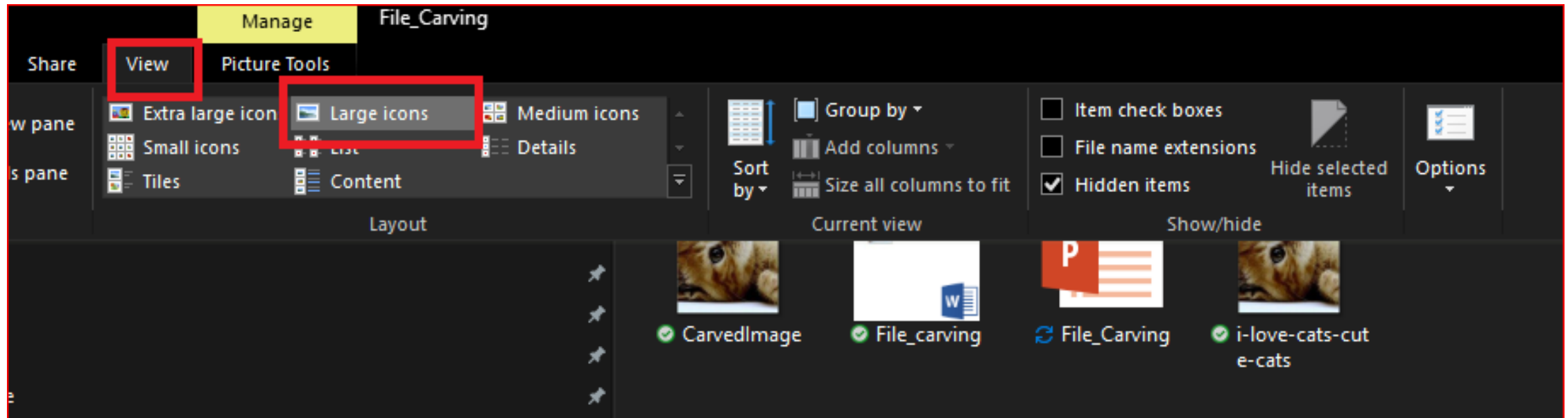
FIND FILE SIGNATURE OF WORD, JPG, AND PPT

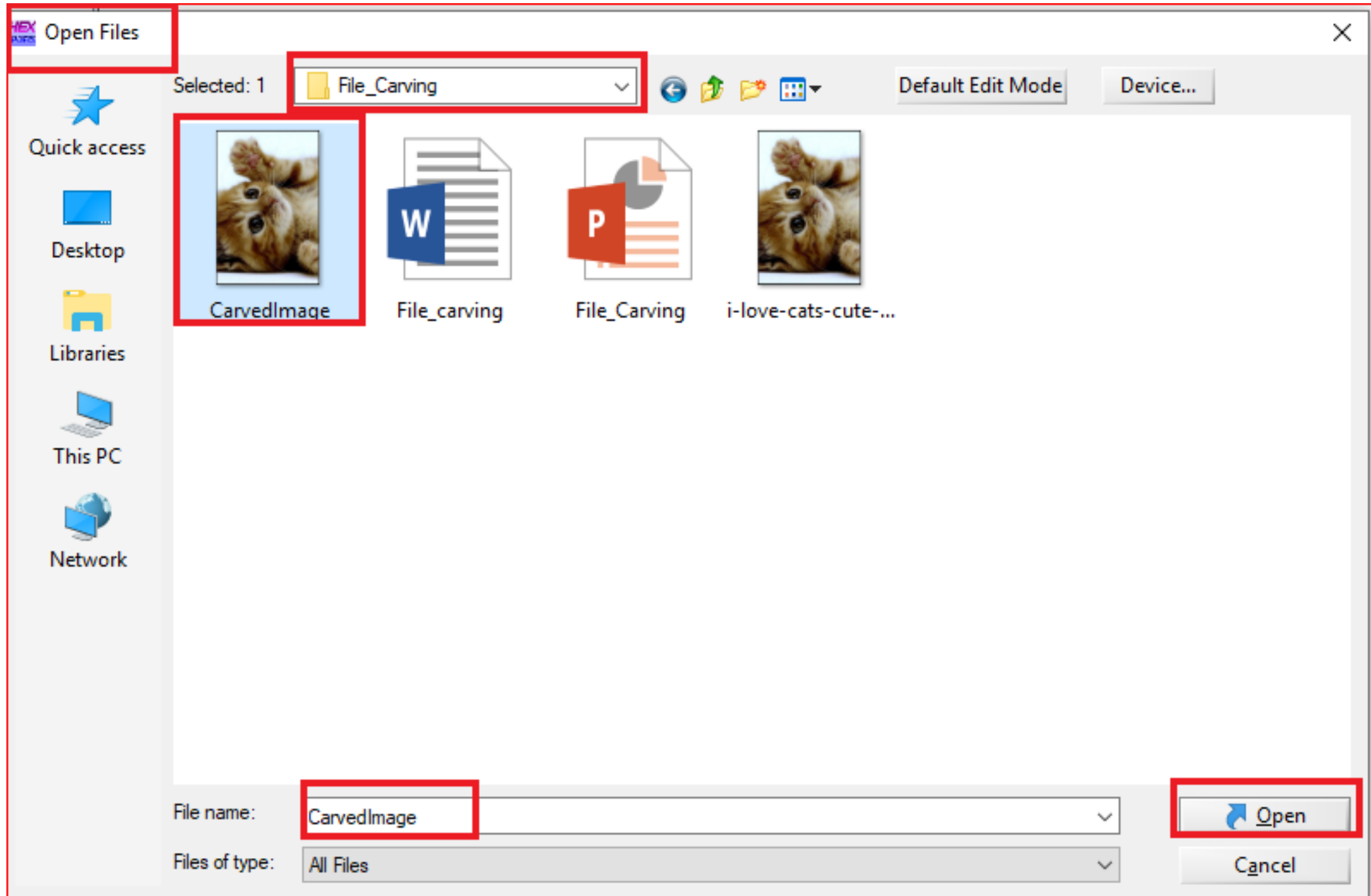
FSCS727_Forensics > Labs > File_Carving

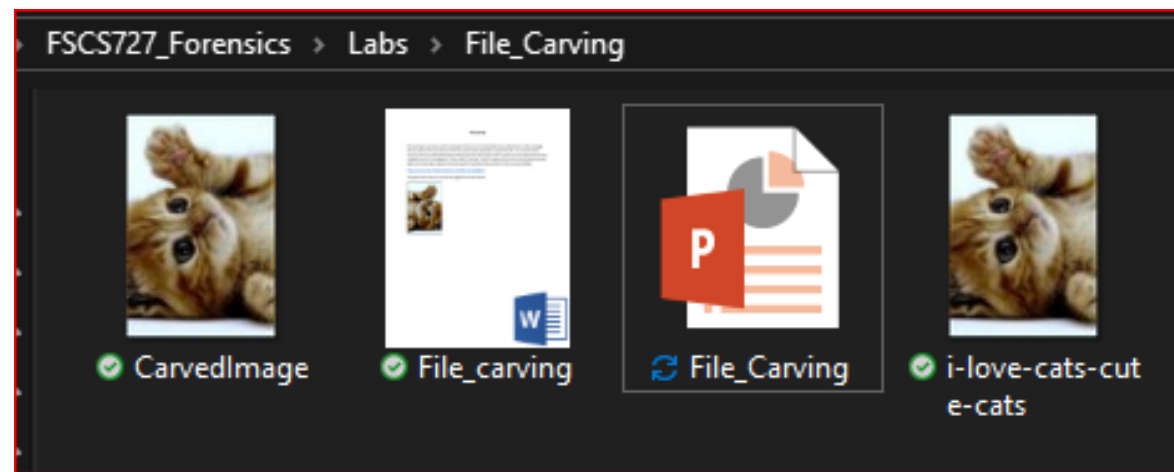
Name	Status	Date	Type	Size	Tags
 CarvedImage	✓	6/30/2018 2:16 PM	JPG File	84 KB	
 File_carving	✓	6/30/2018 1:45 PM	Microsoft Word D...	96 KB	
 File_Carving	↻	10/12/2018 10:35 PM	Microsoft PowerP...	2,039 KB	
 i-love-cats-cute-cats	✓	6/30/2018 1:41 PM	JPG File	98 KB	



VIEW LARGE ICONS



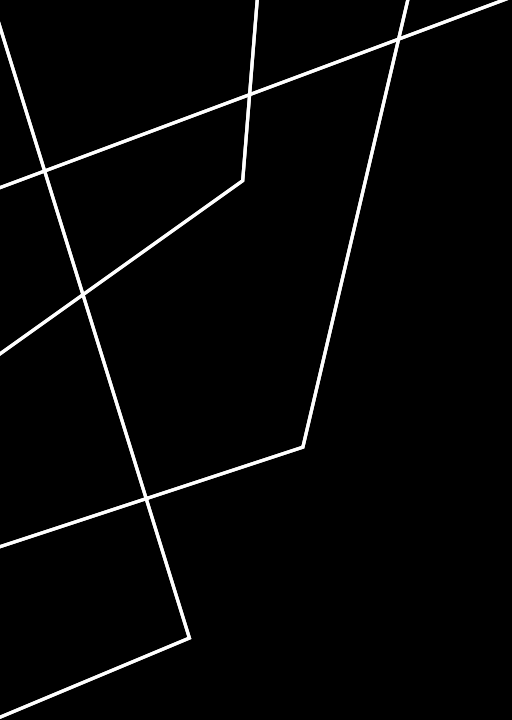




CarvedImage.jpg																		
Offset	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	ANSI	ASCII
00000000	FF	D8	FF	E0	00	10	4A	46	49	46	00	01	01	01	00	DC	ÿøÿà	JFIF Û
00000016	00	DC	00	00	FF	DB	00	43	00	02	01	01	01	01	01	02	Û ŷÛ	C

CarvedImage.jpg File_carving.docx																	
Offset	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	
00000000	50	4B	03	04	14	00	06	00	08	00	00	00	21	00	1C	41	EK
00000016	A8	2E	66	01	00	00	54	05	00	00	13	00	08	02	5B	43	..

CarvedImage.jpg		File_carving.docx				File_Carving.pptx												
Offset		0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	
00000000		50	4B	03	04	14	00	06	00	08	00	00	00	21	00	93	1B	PK
00000016		DB	FD	2A	02	00	00	DE	14	00	00	13	00	08	02	5B	43	Ûý*



EXTRACT THE JPG
FILE FROM AN
UNKNOWN FILE

TOOLS

- MacOS: Hex Fiend

<https://hexfiend.com/>

- Windows: Frhed

<https://sourceforge.net/projects/frhed/>

FSCS727_Forensics > Labs > File_Carving

The screenshot shows the 'File_Carving' lab interface. At the top, the breadcrumb navigation reads 'FSCS727_Forensics > Labs > File_Carving'. Below this, there are three file type options, each with a green checkmark icon and a label: 'CarvedImage' (represented by a cat image), 'File_carving' (represented by a document icon with a red box around it), and 'File_Carving' (represented by a presentation icon). The 'File_carving' option is the one selected for the lab.



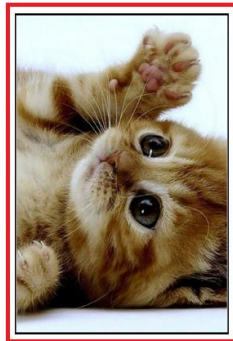
GOAL OF THE LAB

File carving

File carving is a process used in computer forensics to extract data from a disk drive or other storage device without the assistance of the file system that originally created the file. It is a method that recovers files at unallocated space without any file information and is used to recover data and execute a digital forensic investigation. It also called "carving," which is a general term for extracting structured data out of raw data, based on format specific characteristics present in the structured data.

<https://resources.infosecinstitute.com/file-carving/#gref>

The goal of the lab is to extract the jpg file from the Word.



our goal: extract
the image from
clusters!

[illegible]

FILE HEADER OFFSET – 0F5E

File_carving.docx

File_carving.docx

Offset	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	ANSI	ASCII
00000EF0	F2	DD	5C	6B	B7	D5	2D	16	CE	77	BB	E3	84	AE	2C	46	òÝ\k-Ö- Îw»ã,,@, F	
00000F00	FD	03	26	AC	F1	C3	28	9A	97	46	91	22	A4	F2	4C	21	ý &-ñÃ(š-F'"«òL!	
00000F10	80	50	54	F7	7B	69	82	89	BE	19	DE	D5	E4	85	EE	2F	€PT÷{i,%¼ BÖä...i/	
00000F20	E8	D1	FF	00	00	00	FF	FF	03	00	50	4B	03	04	0A	00	èÑÿ ÿÿ PK	
00000F30	00	00	00	00	00	00	21	00	57	80	15	44	36	4C	01	00	! W€ D6L	
00000F40	36	4C	01	00	16	00	00	00	77	6F	72	64	2F	6D	65	64	6L word/med	
00000F50	69	61	2F	69	6D	61	67	65	31	2E	6A	70	65	67	FF D8		ia/image1.jpegÿØ	
00000F60	FF	E0	00	10	4A	46	49	46	00	01	01	01	00	DC	00	DC	ÿà JFIF Û Ü	
00000F70	00	00	FF	DB	00	43	00	02	01	01	01	01	01	02	01	01	ÿÛ C	
00000F80	01	02	02	02	02	02	04	03	02	02	02	02	05	04	04	03		
00000F90	04	06	05	06	06	06	05	06	06	06	07	09	08	06	07	09		
00000FA0	07	06	06	08	0B	08	09	0A	0A	0A	0A	0A	06	08	0B	0C		
00000FB0	0B	0A	0C	09	0A	0A	0A	FF	DB	00	43	01	02	02	02	02		
00000FC0	02	02	05	03	03	05	0A	07	06	07	0A	0A	0A	0A	0A	0A		
00000FD0	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A		
00000FE0	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A		
00000FF0	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	FF	C0	00	11		
00001000	08	02	1C	01	68	03	01	22	00	02	11	01	03	11	01	FF		
00001010	C4	00	1F	00	00	01	05	01	01	01	01	01	01	00	00	00		
00001020	00	00	00	00	00	01	02	03	04	05	06	07	08	09	0A	0B		
00001030	FF	C4	00	B5	10	00	02	01	03	03	02	04	03	05	05	04		
00001040	04	00	00	01	7D	01	02	03	00	04	11	05	12	21	31	41		
00001050	06	13	51	61	07	22	71	14	32	81	91	A1	08	23	42	B1		
00001060	C1	15	52	D1	F0	24	33	62	72	82	09	0A	16	17	18	19		
00001070	1A	25	26	27	28	29	2A	34	35	36	37	38	39	3A	43	44		
00001080	45	46	47	48	49	4A	53	54	55	56	57	58	59	5A	63	64		
00001090	65	66	67	68	69	6A	73	74	75	76	77	78	79	7A	83	84		

File_carving.docx
C:\Users\xuwe524\...

File size:
DOS name:
Default Edit Mode
State:

Find Hex Values

The following hex values will be searched:

FFD8FFE0

☐ Use as wildcard: 3F

Search: All

☐ Cond.: offset mod 512 = 0

☐ Search in block only

☐ Search in all open windows

☒ List search hits, up to 10000

OK Cancel Help

FILE TRAILER OFFSET – 15B92

Offset	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
00015B50	34	F0	EF	FE	9E	AC	E8	AF	E5	D7	C5	9F	B4	1F	C7	AF
00015B60	1D	F8	72	4F	0C	F8	EB	E3	67	8B	B5	BD	36	E3	6F	9F
00015B70	A7	EA	DE	24	BA	B8	82	42	AF	B9	4B	23	C8	55	88	65
00015B80	04	64	1C	10	28	AE	5A	B5	39	E7	73	BF	0F	47	D9	D3
00015B90	B5	CF	FF	D9	50											
00015BA0	21	00	AA	52	25											
00015BB0	00	00	77	6F	72											
00015BC0	6D	65	31	2E	78											
00015BD0	FA	1F	C4	DC	1D											
00015BE0	9B	84	EC	26	25											
00015BF0	13	02	25	39	16											
00015C00	09	F4	92	FE	9A											
00015C10	96	59	DA	6C	60											
00015C20	8F	34	9E	CB	57											
00015C30	A9	E2	00	94	06											
00015C40	70	01	D3	10	12											
00015C50	97	E1	8E	88	51											
00015C60	9D	72	99	07	B2											
00015C70	42	56	59	54	0E											
00015C80	02	71	EA	80	14											
00015C90	9D	DD	C2	F9	80											

Find Hex Values

The following hex values will be searched:

FFD9

☐ Use as wildcard: 3F

Search: All

☐ Cond.: offset mod 512 = 0

☐ Search in block only

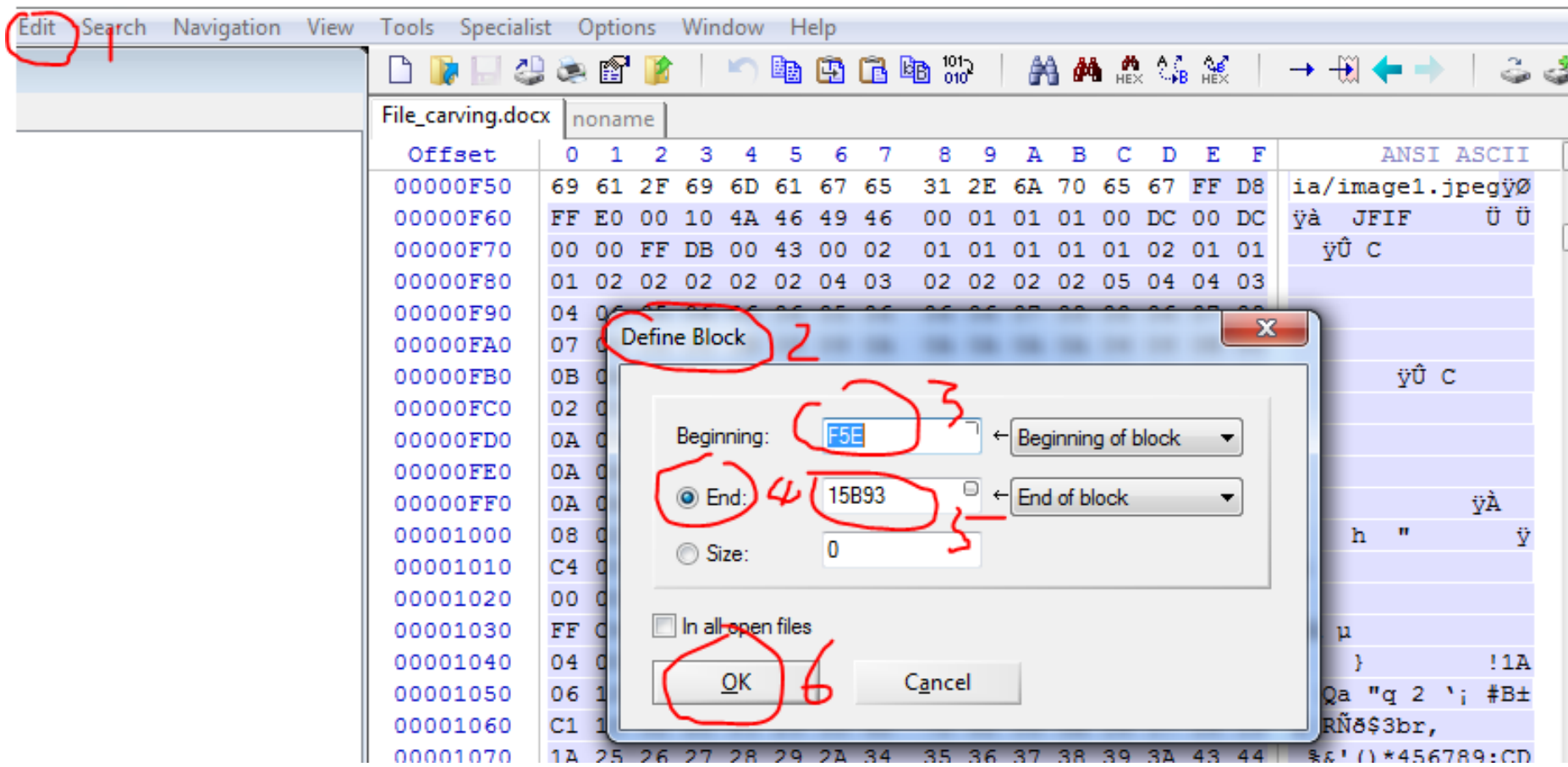
☐ Search in all open windows

☐ List search hits, up to 10000

OK Cancel Help

FILE HEADER OFFSET – 0F5E

FILE TRAILER OFFSET – 15B93



EDIT->COPY BLOCK

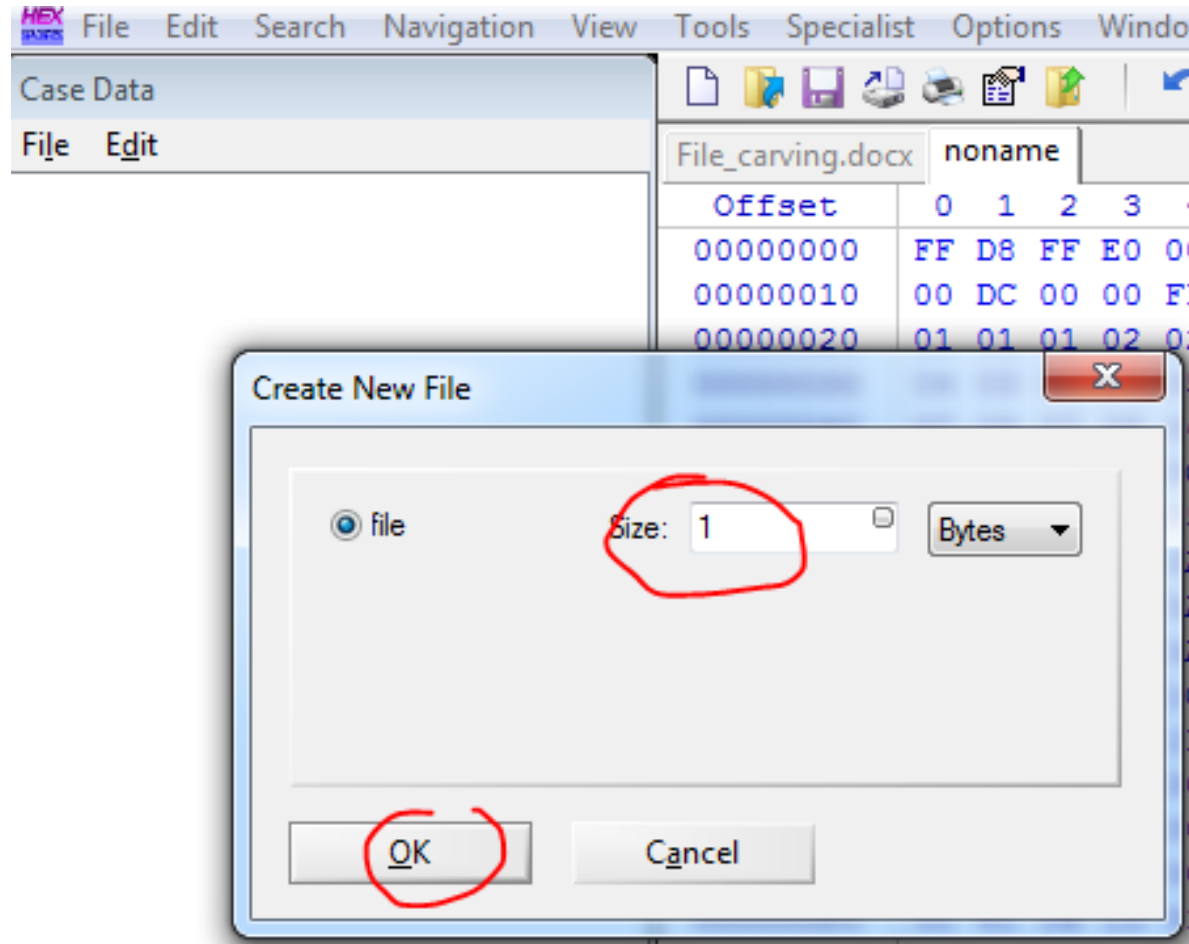
Search Navigation View Tools Specialist Options Window Help

File_carving.docx

noname

Offset	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	ANSI ASCII
00000F50	69	61	2F	69	6D	61	67	65	31	2E	6A	70	65	67	FF	D8	ia/image1.jpegÿ0
00000F60	FF	E0	00	10	4A	46	49	46	00	01	01	01	00	DC	00	DC	ÿà JFIF ÿÿ
00000F70	00	00	FF	DB	00	43	00	02	01	01	01	01	01	02	01	01	ÿÿ C
00000F80	01	02	02	02	02	04	03	02	02	02	02	05	04	04	03		
00000F90	04	06	05	06	06	06	05	06	06	06	07	09	08	06	07	09	
00000FA0	07	06	06	08	0B	08	09	0A	0A	0A	0A	0A	06	08	0B	0C	
00000FB0	0B	0A	0C	09	0A	0A	0A	FF	DB	00	43	01	02	02	02	02	ÿÿ C
00000FC0	02	02	05	03	03	05	0A	07	06	07	0A	0A	0A	0A	0A	0A	
00000FD0	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	
00000FE0	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	
00000FF0	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	FF	C0	00	11	ÿÀ
00001000	08	02	1C	01	68	03	01	22	00	02	11	01	03	11	01	FF	h " ÿ
00001010	C4	00	1F	00	00	01	05	01	01	01	01	01	01	00	00	00	Ä
00001020	00	00	00	00	00	01	02	03	04	05	06	07	08	09	0A	0B	
00001030	FF	C4	00	B5	10	00	02	01	03	03	02	04	03	05	05	04	ÿÄ µ
00001040	04	00	00	01	7D	01	02	03	00	04	11	05	12	21	31	41	}
00001050	06	13	51	61	07	22	71	14	32	81	91	A1	08	23	42	B1	Qa "q 2 ' ; #B±
00001060	C1	15	52	D1	F0	24	33	62	72	82	09	0A	16	17	18	19	Á RÑø\$3br,
00001070	1A	25	26	27	28	29	2A	34	35	36	37	38	39	3A	43	44	±&'()*456789:CD
00001080	45	46	47	48	49	4A	53	54	55	56	57	58	59	5A	63	64	EFGHIJSTUVWXYZcd
00001090	65	66	67	68	69	6A	73	74	75	76	77	78	79	7A	83	84	efghijstuvwxyzf,,
000010A0	85	86	87	88	89	8A	92	93	94	95	96	97	98	99	9A	A2	...t+^%\$'""*~¡¢£„
000010B0	A3	A4	A5	A6	A7	A8	A9	AA	B2	B3	B4	B5	B6	B7	B8	B9	£¥¦§¨ª«¬®¯°±²³´µ¶·¸¹
000010C0	BA	C2	C3	C4	C5	C6	C7	C8	C9	CA	D2	D3	D4	D5	D6	D7	º»¼½¾¿ÀÁÂÃÄÅÆÇÈÉÊËÌÍÎÏÐ
000010D0	D8	D9	DA	E1	E2	E3	E4	E5	E6	E7	E8	E9	EA	F1	F2	F3	ÒÓÔÕÖ×ØÙÚÛÜÝÞßàáâãäåæçèéêëìíî
000010E0	F4	F5	F6	F7	F8	F9	FA	FF	C4	00	1F	01	00	03	01	01	ðñòó÷øùúÿÀ
000010F0	01	01	01	01	01	01	01	00	00	00	00	00	00	01	02	03	
00001100	04	05	06	07	08	09	0A	0B	FF	C4	00	B5	11	00	02	01	ÿÀ µ
00001110	02	04	04	03	04	07	05	04	04	00	01	02	77	00	01	02	w
00001120	03	11	04	05	21	31	06	12	41	51	07	11	71	13	22	32	!1 Aq aq "2
00001130	81	08	14	42	91	A1	B1	C1	09	23	33	52	F0	15	62	72	R':+Á #3RA br

PASTE TO A NEW FILE

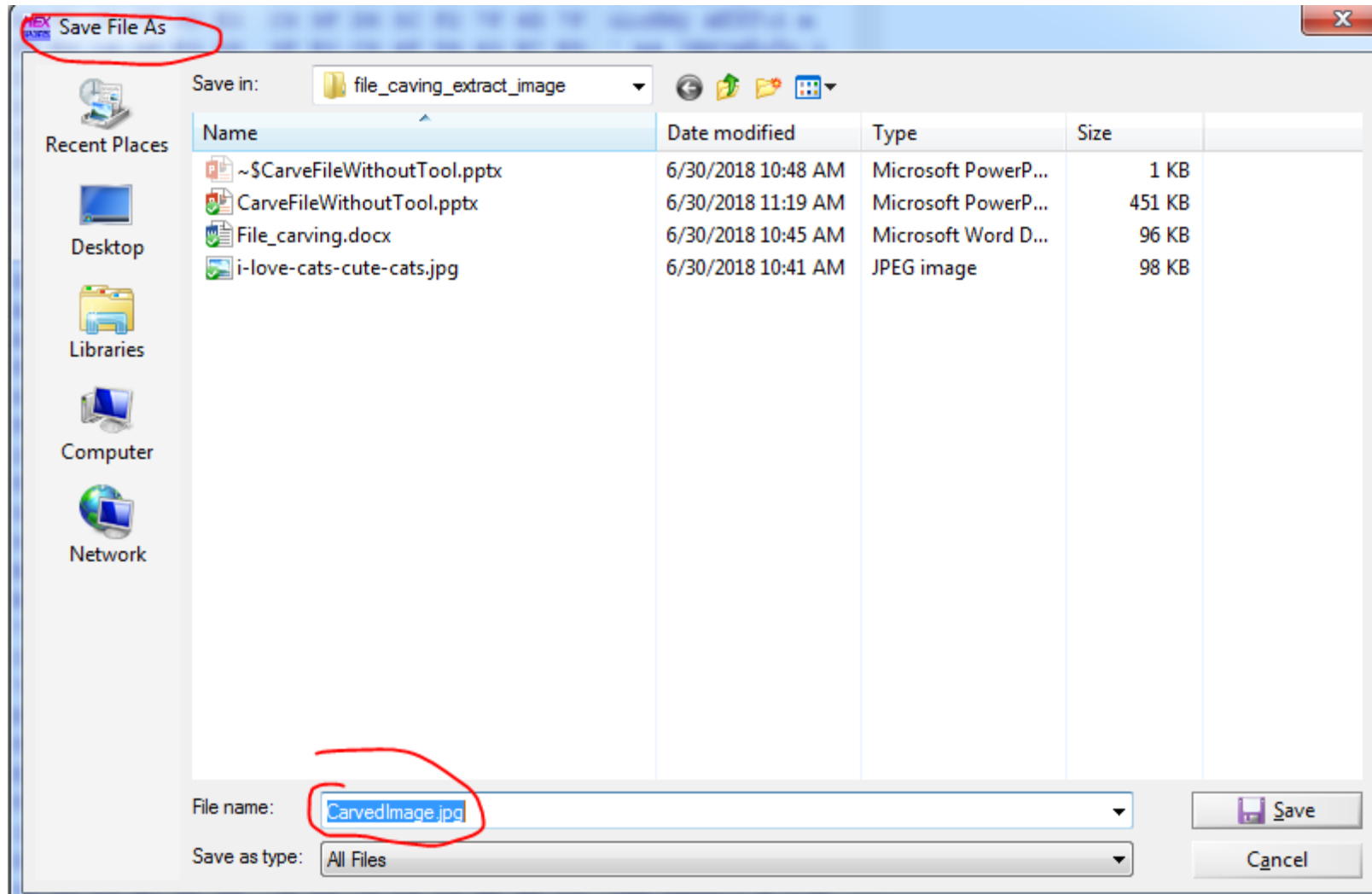


FILE->

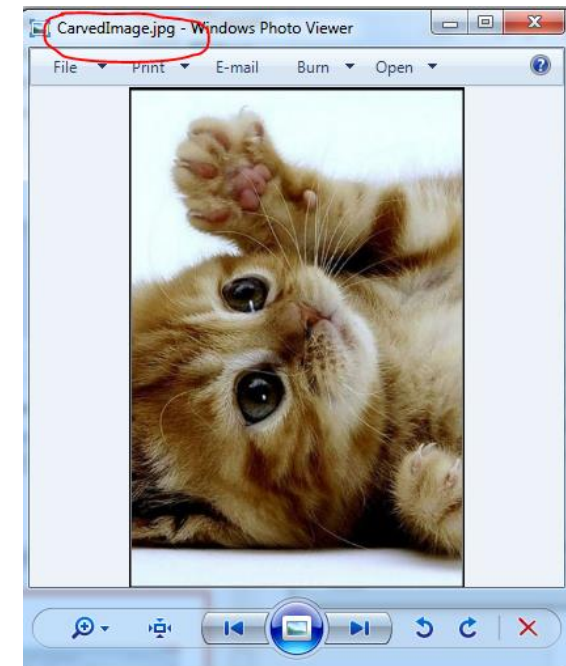
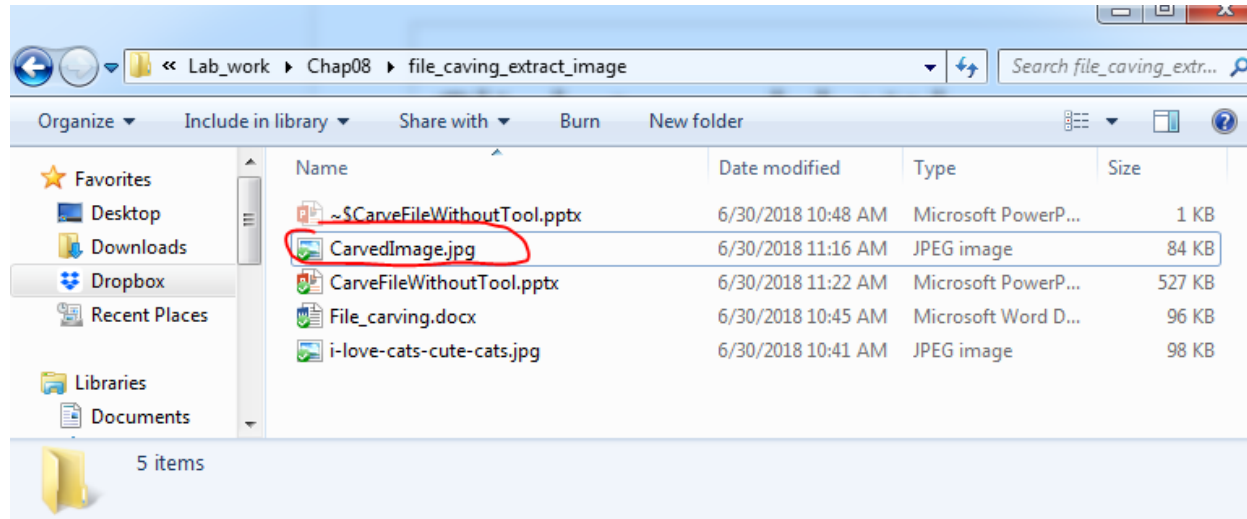
File_carving.docx noname new

Offset	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	ANSI	ASCII
00000000	FF	D8	FF	E0	00	10	4A	46	49	46	00	01	01	01	00	DC	ÿøÿà	JFIF Û
00000010	00	DC	00	00	FF	DB	00	43	00	02	01	01	01	01	01	02	Û	ÿô c
00000020	01	01	01	02	02	02	02	02	04	03	02	02	02	02	05	04		
00000030	04	03	04	06	05	06	06	06	05	06	06	06	07	09	08	06		
00000040	07	09	07	06	06	08	0B	08	09	0A	0A	0A	0A	0A	06	08		
00000050	0B	0C	0B	0A	0C	09	0A	0A	0A	FF	DB	00	43	01	02	02		ÿô c
00000060	02	02	02	02	05	03	03	05	0A	07	06	07	0A	0A	0A	0A		
00000070	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A		
00000080	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A		
00000090	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	0A	FF	C0		ÿÀ
000000A0	00	11	08	02	1C	01	68	03	01	22	00	02	11	01	03	11		h "
000000B0	01	FF	C4	00	1F	00	00	01	05	01	01	01	01	01	01	00		ÿÃ
000000C0	00	00	00	00	00	00	00	01	02	03	04	05	06	07	08	09		
000000D0	0A	0B	FF	C4	00	B5	10	00	02	01	03	03	02	04	03	05		ÿÃ p
000000E0	05	04	04	00	00	01	7D	01	02	03	00	04	11	05	12	21		}
000000F0	31	41	06	13	51	61	07	22	71	14	32	81	91	A1	08	23	1A	Qa "q 2 'i #
00000100	42	B1	C1	15	52	D1	F0	24	33	62	72	82	09	0A	16	17	B±Á	RÑð\$3br,
00000110	18	19	1A	25	26	27	28	29	2A	34	35	36	37	38	39	3A		%&'()*+456789:
00000120	43	44	45	46	47	48	49	4A	53	54	55	56	57	58	59	5A		CDEFGHIJSTUVWXYZ
00000130	63	64	65	66	67	68	69	6A	73	74	75	76	77	78	79	7A		cdefghijstuvwxyz
00000140	83	84	85	86	87	88	89	8A	92	93	94	95	96	97	98	99		f„…†+~`%Š'""•—™
00000150	9A	A2	A3	A4	A5	A6	A7	A8	A9	AA	B2	B3	B4	B5	B6	B7		šçŁŒŒ!\$~©~*~*~pŒ·

FILE->SAVE FILE AS

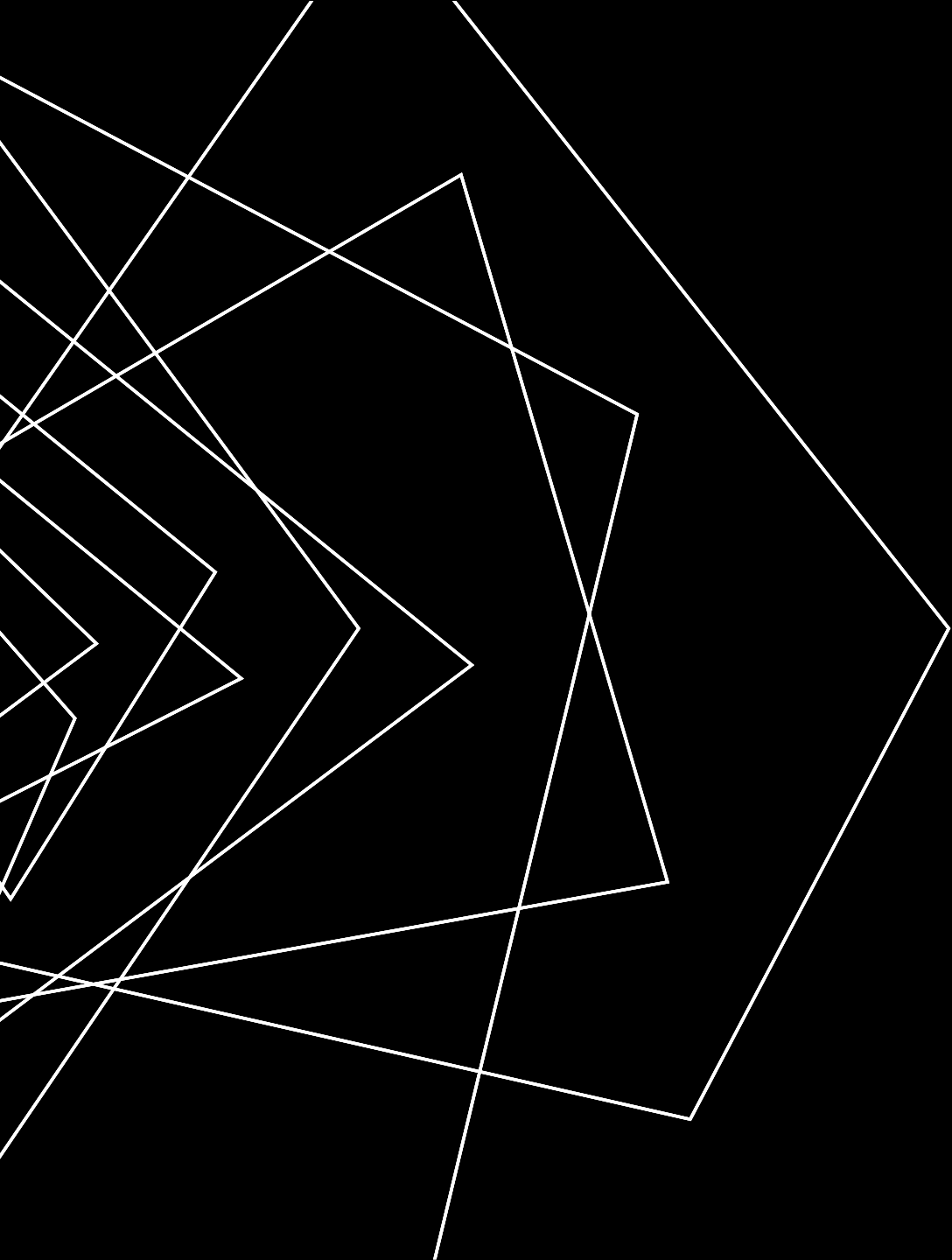


CARVED FILE



USING TOOLS TO RECOVER

- [PhotoRec](#) is open source recovery software designed to recover lost files, including video, documents, and archives from hard disks, CD-ROMs, and lost pictures (thus the “photo recovery” name) from digital camera memory.
- PhotoRec ignores the file system and goes after the underlying data, so it will still work even if your media’s file system has been severely damaged or reformatted.
- It is available for the Windows, Linux, and MAC operating systems



THANK YOU

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ACB1 Room 335